

# CURRICULUM VITAE

## Andrzej Hryczuk

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|-----------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Date of birth:  | 11 <sup>th</sup> December 1983                                                        | Address:                                                                                                             |
| Place of birth: | Warsaw, Poland                                                                        | NCBJ                                                                                                                 |
| Nationality:    | Polish                                                                                | ul. Pasteura 7,                                                                                                      |
| Languages:      | English (C2), Russian (B1),<br>German (B1), Italian (A1),<br>Polish ( <i>native</i> ) | 02-093 Warsaw, Poland<br><a href="https://github.com/ahryczuk">ahryczuk.github.io</a><br>andrzej.hryczuk@ncbj.gov.pl |

### Biosketch

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Theoretical physicist working on particle cosmology and high energy physics. Research interests include: dark matter, beyond SM phenomenology and QFT at finite temperature. Did extensive research on all these topics and currently focused on evolution of dark matter in the Early Universe.

H-index: 17 (according to [INSPIRE](#) database).

### Work experience

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|---------------------|------------------------------------------------------------------------------------------------------------------------|
| February 2026 -     | Associate Editor (part-time), <b>Physical Review D</b> ,                                                               |
| June 2025 -         | Associate Professor, <b>National Centre for Nuclear Research</b> , Warsaw, Poland                                      |
| Dec 2018 - May 2025 | Assistant Professor ( <i>Adiunkt</i> ), <b>National Centre for Nuclear Research</b> , Warsaw, Poland                   |
| Dec 2015 - Nov 2018 | Postdoctoral Fellow, <b>University of Oslo</b> , Norway                                                                |
| Sep 2015 - Dec 2015 | Postdoctoral Fellow, <b>National Centre for Nuclear Research</b> , Warsaw, Poland                                      |
| Oct 2012 - Sep 2015 | Postdoctoral Fellow, <b>Technische Universität München</b> , Munich, Germany                                           |
| Nov 2008 - Oct 2010 | Early Stage Researcher, Marie Curie Research Training Network "UniverseNet",<br><b>SISSA and INFN</b> , Trieste, Italy |
| Jul - Oct 2006      | Internship at <b>CERN</b> in the OSQAR Collaboration, Geneva, Switzerland<br>Group of Dr. Andrzej Siemko               |

### Education

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|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023        | Habilitation at National Centre for Nuclear Research<br>Title: <i>Accurate determination of the dark matter relic abundance</i>                                              |
| 2008 - 2012 | Ph.D. Student at SISSA, Trieste, Italy<br>Thesis title: <i>Loop and non-perturbative effects in the dark matter phenomenology</i><br>Supervisor: Prof. Piero Ullio           |
| 2007 - 2008 | Ph.D. Student at the Faculty of Physics, Warsaw University, Poland                                                                                                           |
| 2002 - 2007 | M.Sc. Degree in Theoretical Physics with Honours, Warsaw University<br>Thesis title: <i>Axions in Particle Physics and Cosmology</i><br>Supervisor: Prof. Krzysztof Meissner |
| 1998 - 2002 | Matriculation with Honours, The 14 <sup>th</sup> Secondary School in Warsaw<br>with the prize of the Mayor of Warsaw-Center District                                         |

## Research grants

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep 2022-Aug 2027 | PI of SONATA BIS grant 2021/42/E/ST2/00009, National Science Centre, Poland<br>Project title: <i>Dark matter and baryogenesis within multicomponent dark sectors and nonstandard cosmological models</i>        |
| Jul 2019-Jul 2023 | PI of SONATA grant 2018/31/D/ST2/00813, National Science Centre, Poland<br>Project title: <i>The impact of non-equilibrium and thermal effects on the evolution of Dark Matter in the Early Universe plasma</i> |

## Scholarships and Awards

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|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2017                                         | PRD Editors' Suggestion of paper "Early kinetic decoupling of dark matter: when the standard way of calculating the thermal relic density fails"        |
| Mar 2008                                     | Award for the best M.Sc. thesis at the Faculty of Physics, Warsaw University                                                                            |
| Oct 2004 - Oct 2007                          | Scholarship of the Minister of Science and Higher Education                                                                                             |
| 2 <sup>nd</sup> Oct 2006                     | Award of the Marshal of the Masovian Voivodeship                                                                                                        |
| 18 <sup>th</sup> Dec 2004                    | Award for the best article published in the periodical Delta in academic year 2002/2003: <i>Niebieski las</i> (Blue Forest), Delta, <b>351</b> (2003) 4 |
| 23 <sup>rd</sup> - 30 <sup>th</sup> May 2002 | 1 <sup>st</sup> Prize at the 15 <sup>th</sup> IYPT, Odessa, Ukraine                                                                                     |
| 22 <sup>nd</sup> - 29 <sup>th</sup> May 2001 | 3 <sup>rd</sup> Prize at the 14 <sup>th</sup> IYPT, Espoo, Finland                                                                                      |

## Teaching and public outreach

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| <b>Teaching</b>            | <i>Classical Electrodynamics</i> , Lectures (NCBJ, winter term 2025/26)<br><i>Selected Topics in Dark Matter</i> , open Lecture Series (UiO, spring 2018)<br><i>Classical Mechanics and Special Relativity</i> , Tutor (Warsaw University, spring 2019)<br><i>Statistical Physics and Thermodynamics</i> , Master tutor (TUM, 2013/2014)<br><i>Mathematical Analysis I with Algebra</i> , Tutor (Warsaw University, 2007/2008)                                                                         |
| <b>Ph.D Supervision</b>    | supervision of Esau Cervantes<br>National Centre for Nuclear Research, 2022-<br><br>day-to-day co-supervision of Francesco Dighera,<br>Technische Universität München, 2014-2015<br>Thesis title: <i>Radiative Corrections in Relic Density Calculations</i>                                                                                                                                                                                                                                           |
| <b>M.Sc. Supervision</b>   | co-supervision of August Geelmuyden, Thesis title: <i>On the Prospects of Microlensing WIMPY Halos</i> , University of Oslo, 2017-2018, Thesis grade: A<br><br>co-supervision of Alessio Pizzini, Thesis title: <i>Thermal production of self-interacting dark matter</i> , University of Oslo, 2017-2018, Thesis grade: C<br><br>co-supervision of Ask Markestad, Thesis title: <i>Dark Matter Bound State Formation for Pseudo-Scalar Mediators</i> , University of Oslo, 2016-2017, Thesis grade: A |
| <b>Trainee Supervision</b> | supervision of Adam Gomulka, M.Sc. student at Warsaw University, July 2024                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Popular talks

panelist in a discussion 'Im mniej wiem, tym lepiej śpię?' (*The less I know, the better I sleep?*),

Skoczkowo, Poland (4<sup>th</sup> Aug 2018)

Interview in the *Radio Naukowe* podcast, (online) (26<sup>th</sup> Jul 2024)

'Dlaczego Wszechświat się rozszerza?' (*Why does the Universe expand?*),

Letnie Spotkania z Nauką, (online) (16<sup>th</sup> Jul 2022)

Interview in the Polish Radio program [EUREKA](#)  
(23<sup>rd</sup> Nov 2021)

'Ciemna materia okiem fizyka cząstek' (*Dark matter seen by a particle physicist*),  
DM day 2021, (online) (28<sup>th</sup> Oct 2021)

'Ciemność widzę, ciemność... czyli jak zobaczyć ciemną materię' (*How to see the dark matter*),

Maraton wykładowy z Delta, (online) (25<sup>th</sup> Feb 2021)

'Ciemna strona Wszechświata' (*The dark side of the Universe*),  
Festiwal Nauki, Warsaw, Poland (26<sup>th</sup> Sep 2020)

'Co kryje się we wnętrzu czarnej dziury?' (*What is inside a black hole?*),  
Letnie Spotkania z Nauką, Skoczkowo, Poland (13<sup>th</sup> Jul 2019)

'Entropia i strzałka czasu' (*Entropy and the arrow of time*),  
Letnie Spotkania z Nauką, Skoczkowo, Poland (4<sup>th</sup> Aug 2018)

'Z czego zbudowany jest Wszechświat?' (*What is the Universe Made of?*),  
Letnie Spotkania z Nauką, Skoczkowo, Poland (17<sup>th</sup> Aug 2013)

'The International Young Physicist Tournament', Polish Seminar on Physics  
Teaching, Warsaw, Poland (19<sup>th</sup> Mar 2003)

## Popular papers

*Płeczek w szklance wody* (*Plastic Ball in a Glass of Water*)

A. Hryczuk, R. Żak, *Delta*, **353** (2003) 10

*Niebieski las* (*Blue Forest*)

A. Hryczuk, R. Żak, *Delta*, **351** (2003) 4

*O unoszeniu się ciał na wodzie* (*On Floating of Bodies*)

A. Hryczuk, R. Żak, *Delta*, **347** (2003) 11

## Invited Talks

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28<sup>th</sup> May 2026 'Making Dark Matter Cool Again: dynamics of self-interacting dark sectors', invited seminar at University of Warsaw, Warsaw, Poland

8<sup>th</sup> May 2026 'A case for calculating dark matter momentum distribution ', seminar at New York University, USA

14<sup>th</sup> Jan 2026 'Freeze-in & Freeze-out production of Two-component Dark Sectors at the phase space level', invited talk at HECAP workshop, NYU Abu Dhabi, UAE

20<sup>th</sup> Nov 2025 'DRAKE2', invited talk at The Dark Universe in Phase Space Focus Week, IFPU Trieste, Italy

22<sup>nd</sup> Sep 2025 'Sequential Freeze-in: A Tale of Two Scalars', invited plenary talk at the Scalars 2025 conference, Warsaw, Poland

08<sup>th</sup> Jan 2025 'Particle Dark Matter: nature and origins', invited colloquium at the Centre for Theoretical Physics PAN, Warsaw, Poland

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|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 04 <sup>th</sup> Apr 2024 | <i>'Towards robust predictions for thermal production of multicomponent dark matter'</i> ,<br>Invited seminar at the Faculty of Physics University of Warsaw, Warsaw, Poland |
| 27 <sup>th</sup> Sep 2023 | <i>'Non-standard Dark Matter Freeze-out'</i> , Invited talk at the 3rd DMNet International Symposium, Padova, Italy                                                          |
| 15 <sup>th</sup> Feb 2023 | <i>'Elastic Self-scatterings in the Calculation of Dark Matter Relic Abundance'</i> , invited talk at the Particle Astrophysics in Poland meeting, Krakow, Poland            |
| 22 <sup>nd</sup> Oct 2022 | <i>'Advances in Dark Matter Production Theory'</i> , invited talk at the Symposium of Polish Physical Society, Katowice, Poland                                              |
| 16 <sup>th</sup> Sep 2022 | <i>'Relic Abundance Theory: New Developments'</i> , invited talk at the DM3 RIKEN workshop, Kobe, Japan (online)                                                             |
| 1 <sup>st</sup> Apr 2022  | <i>'Developments in Dark matter Relic Abundance calculations'</i> , invited seminar at Institute of Mathematical Sciences, Chennai, India (online)                           |
| 25 <sup>th</sup> Nov 2021 | <i>'Freeze-in of Dark Matter: recent developments'</i> , invited seminar at University of Warsaw, Warsaw, Poland (online)                                                    |
| 23 <sup>rd</sup> Sep 2021 | <i>'Dark Matter and the <math>H_0</math> tension'</i> , invited parallel talk at the Polish Physicists' Summit (Zjazd Fizyków Polskich), Bydgoszcz, Poland (online)          |
| 9 <sup>th</sup> Jan 2020  | <i>Heavy WIMPs: status and future prospects</i> , invited plenary talk at Rencontres du Vietnam, Quy Nhon,                                                                   |
| 24 <sup>th</sup> Oct 2019 | <i>'Dark Matter at the TeV frontier'</i> , invited seminar at University of Warsaw, Warsaw, Poland                                                                           |
| 14 <sup>th</sup> Sep 2019 | <i>'Non-equilibrium effects in the evolution of Dark Matter'</i> , invited parallel talk at the Polish Physicists' Summit (Zjazd Fizyków Polskich), Cracow, Poland           |
| 18 <sup>th</sup> Mar 2019 | <i>'Non-equilibrium effects in the evolution of Dark Matter'</i> , invited joint CPT/CPPM seminar, Marseille, France                                                         |
| 9 <sup>th</sup> Nov 2018  | <i>'Particle Dark Matter at a Crossroads'</i> , invited colloquium at Institute of Theoretical Astrophysics, Oslo, Norway                                                    |
| 22 <sup>th</sup> Dec 2017 | <i>'Kinetic Decoupling of Dark Matter and its impact on the Relic Density'</i> , invited seminar at University of Warsaw, Warsaw, Poland                                     |
| 7 <sup>th</sup> Nov 2017  | <i>'(More) exceptions in the calculations of relic abundances'</i> , invited seminar at LPTHE, Paris, France                                                                 |
| 28 <sup>th</sup> Sep 2016 | <i>'Relic Density at NLO: IR finiteness'</i> , invited talk at "Rethinking Quantum Field Theory" workshop, DESY, Hamburg, Germany                                            |

## Contributing talks and presentations

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| 1 <sup>st</sup> Nov 2024   | <i>'Towards robust predictions for thermal production of multicomponent dark matter'</i> , Kashiwa-no-ha Symposium, Tokyo, Japan                                                     |
| 24 <sup>th</sup> Oct 2024  | <i>'Freezing-in Cannibal Dark Sectors'</i> , COSMO-24, Kyoto, Japan                                                                                                                  |
| 29 <sup>th</sup> Jun 2023  | <i>'Dark matter production out of kinetic equilibrium: the latest developments '</i> , Contributing talk at the PASCOS conference, Irvine CA, USA                                    |
| 28 <sup>th</sup> Sep 2022  | <i>'Dark matter freeze-out and freeze-in beyond kinetic equilibrium'</i> , NuDM 2022, Sharm El Sheikh, Egypt                                                                         |
| 1 <sup>st</sup> May 2022   | <i>'Dark Matter Freeze-out and Freeze-in Beyond Kinetic Equilibrium'</i> , SUSY 2022, Ioannina, Greece                                                                               |
| 26 <sup>th</sup> May 2022  | <i>'Impact of Self-scattering on Dark Matter Relic Density'</i> , 33rd Rencontres de Blois, France                                                                                   |
| 28 <sup>th</sup> July 2021 | <i>'DRAKE: Dark matter relic abundance beyond kinetic equilibrium'</i> , EPS-HEP conference, DESY, Germany (online)                                                                  |
| 19 <sup>th</sup> May 2021  | <i>'Dark matter relic abundance beyond kinetic equilibrium'</i> , PPC workshop, University of Oklahoma (online)                                                                      |
| 4 <sup>th</sup> Dec 2020   | <i>'Self-interacting DM and the <math>H_0</math> tension'</i> , SAFIR DM workshop, Sao Paolo, Brazil (online)                                                                        |
| 28 <sup>th</sup> Aug 2018  | <i>'Early Kinetic Decoupling of DM: when the standard way of calculating the thermal relic density fails'</i> , COSMO 2018, Deajeon, South Korea                                     |
| 14 <sup>th</sup> Mar 2018  | <i>'Dark Matter Relic Density: revisited'</i> , Rencontres de Moriond, La Thuille, Italy                                                                                             |
| 29 <sup>th</sup> Jul 2016  | <i>'Relic Density at NLO: thermal corrections'</i> , The Dark Side of the Universe, Bergen, Norway                                                                                   |
| 20 <sup>th</sup> Jul 2016  | <i>'Neutralino dark matter at TeV scale revisited'</i> , IDM, Sheffield, UK                                                                                                          |
| 4 <sup>th</sup> Jun 2016   | <i>'Spectral features in the MeV-gap'</i> , Warsaw Workshop on Non-Standard DM, Poland                                                                                               |
| 27 <sup>th</sup> Jan 2016  | <i>'Two (more) exceptions in the calculations of relic abundances'</i> , Theory Seminar, University of Oslo, Norway                                                                  |
| 9 <sup>th</sup> Sep 2015   | <i>'The relic density of heavy neutralinos'</i> , COSMO 2015, Warsaw, Poland                                                                                                         |
| 26 <sup>th</sup> Jun 2014  | <i>'Relic density at NLO: the thermal IR divergence'</i> , PASCOS 2014, Warsaw, Poland                                                                                               |
| 10 <sup>th</sup> Oct 2013  | <i>'The indirect detection signals of Wino dark matter'</i> , Workshop on the Future of Dark Matter Astro-Particle Physics: Insights and Perspectives, ICTP, Trieste, Italy (poster) |
| 5 <sup>th</sup> Mar 2013   | <i>'The indirect detection signals of Wino dark matter'</i> , Astrofizyka Cząstek w Polsce, Jagiellonian University, Kraków, Poland (poster)                                         |
| 20 <sup>th</sup> Feb 2013  | <i>'Sommerfeld enhancement'</i> , HAP Dark Matter 2013, Münster University, Germany                                                                                                  |
| 30 <sup>th</sup> May 2012  | <i>'Electroweak and Sommerfeld corrections to the Wino DM annihilation'</i> , PLANCK 2012, University of Warsaw, Poland                                                              |

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| 14 <sup>th</sup> Sep 2010 | 'Sommerfeld enhancements in the MSSM', 4th UniverseNet School, Lecce, Italy                                      |
| 23 <sup>rd</sup> Jun 2010 | 'The Sommerfeld enhancements in the MSSM', FA51 Group Meeting, Laboratori Nazionali di Frascati, Frascati, Italy |

## Others

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- Referee for journals:  
*Physical Review Letters, Physical Review D, Physics Letters B, JHEP, JCAP, European Physical Journal C, Europhysics Letters, Nuclear Physics B, Acta Physica Polonica B, Computer Physics Communications*
- Referee for funding agencies:  
*Fulbright Foundation (Fulbright Graduate Student Award program)*
- Referee for RDN: *1 habilitation, 1 Phd thesis*
- Evaluation committee member: *1 Phd student (chair), 3 Phd students (member), midterm Phd evaluations*
- Recruitment committee member: *6 postdoc, 1 Phd student*
- Scientific secretary for the *Particle Astrophysics in Poland* conference, 20-21 May 2019, Warsaw
- Organization of the *Dark Matter Day 2021* in Poland outreach event, 28 October 2021
- Co-organization of a workshop on "*Selected topics on future directions in particle physics*", STER Programme, NCBJ, 16-18 September 2024
- Co-organization of the *Particle Astrophysics in Poland* conference, 20-22 February 2025, Warsaw
- Co-organization of the *HECA* seminars, academic years 2019/20 - present, joint NCBJ/UW, Warsaw

## Computer skills

- Programming: C/C++, FORTRAN, Wolfram Language (*Mathematica*)
- Web: Html, CSS, basics of PHP – Webmaster of the SISSA High Energy Sector webpage (Oct 2010 - Nov 2011) and of the Ph.D. Students' Council at Faculty of Physics, Warsaw University (Feb - Sep 2008)

# List of Publications

## Andrzej Hryczuk

The publication culture in the field does not distinguish the first or corresponding authors - the author list is alphabetical and journals do not differentiate authors' contributions. Below I divide the publications into main author papers and contributing author papers. Main authorship entails important contributions to all aspects of the project and publication preparation, where it is understood that often there can be more than one main author.

Bibliometrics from [inSPIRE](#):

Total number of citations : 1862

H-index: 17

### Main Author Publications

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- [1] *Exploring non-equilibrium effects in sequential freeze-in*  
S. Chatterjee, A. Hryczuk  
[arXiv:2604.14688](#) [[hep-ph](#)]
- [2] *Freezing-in cannibals with low-reheating temperature*  
N. Bernal, E. Cervantes, K. Deka, A. Hryczuk  
JHEP 09 (2025) 083 [[arXiv:2506.09155](#) [[hep-ph](#)]]
- [3] *Conversions in two-component dark sectors: a phase space level analysis*  
S. Chatterjee, A. Hryczuk  
JHEP 07 (2025) 279 [[arXiv:2502.08725](#) [[hep-ph](#)]]
- [4] *Freezing-in Cannibal Dark Sectors*  
E. Cervantes, A. Hryczuk  
JHEP 11 (2024) 050 [[arXiv:2407.12104](#) [[hep-ph](#)]]
- [5] *Phase transitions and gravitational waves in a model of  $Z_3$  scalar dark matter*  
N. Benincasa, A. Hryczuk, M. Laletin, K. Kannike  
JHEP 02 (2024) 207 [[arXiv:2312.04627](#) [[hep-ph](#)]]
- [6] *Impact of dark matter self-scattering on its relic abundance*  
A. Hryczuk, M. Laletin  
Phys.Rev.D 106 (2022) 2, 023007 [[arXiv:2204.07078](#) [[hep-ph](#)]]
- [7] *Dark matter freeze-in from semi-production*  
A. Hryczuk, M. Laletin  
JHEP 06 (2021) 026 [[arXiv:2104.05684](#) [[hep-ph](#)]]
- [8] *DRAKE: Dark Matter Relic Abundance beyond Kinetic Equilibrium*  
T. Binder, T. Bringmann, M. Gustafsson, A. Hryczuk  
Eur. Phys. J. C 81, 577 (2021) [[arXiv:2103.01944](#) [[hep-ph](#)]]
- [9] *Self-interacting dark matter from late decays and the  $H_0$  tension*  
A. Hryczuk, K. Jodlowski  
Phys.Rev.D 102 (2020) 4 [[arXiv:2006.16139](#) [[hep-ph](#)]]

- [10] *Impact of uncertainties in the halo velocity profile on direct detection of sub-GeV dark matter*  
A. Hryczuk, K. Karukes, L. Roszkowski, M. Talia  
JHEP **2020** (2020) 081 [[arXiv:2001.09156 \[hep-ph\]](#)]
- [11] *Forbidden frozen-in dark matter*  
L. Darne, A. Hryczuk, D. Karamitros, L. Roszkowski  
JHEP **1911** (2019) 159 [[arXiv:1908.05685 \[hep-ph\]](#)]
- [12] *Testing dark matter with Cherenkov light - prospects of H.E.S.S. and CTA for exploring minimal supersymmetry*  
A. Hryczuk, K. Jodkowski, E. Moulin, L. Rinchuso, L. Roszkowski, E. M. Sessolo, S. Trojanowski  
JHEP **1910** (2019) 043 [[arXiv:1905.00315 \[hep-ph\]](#)]
- [13] *Improved bounds on  $\mathbb{Z}_3$  singlet dark matter*  
A. Hektor, A. Hryczuk, K. Kannike  
JHEP **1903** (2019) 204 [[arXiv:1901.08074 \[hep-ph\]](#)]
- [14] *Early kinetic decoupling of dark matter: when the standard way of calculating the thermal relic density fails*  
T. Binder, T. Bringmann, M. Gustafsson, A. Hryczuk  
Phys.Rev D **96** (2017) no.11 115010 [[arXiv:1706.07433 \[hep-ph\]](#)]
- [15] *The last refuge of mixed wino-Higgsino dark matter*  
M. Beneke, A. Bharucha, A. Hryczuk, S. Recksiegel, P. Ruiz-Femenia  
JHEP **1701** (2017) 002 [[arXiv:1611.00804 \[hep-ph\]](#)]
- [16] *Novel Spectral Features in MeV Gamma Rays from Dark Matter*  
T. Bringmann, A. Galea, A. Hryczuk, Ch. Weniger  
Phys.Rev D **95** (2017) no.4 043002 [[arXiv:1610.04613 \[hep-ph\]](#)]
- [17] *Finite-temperature modification of heavy particle decay and dark matter annihilation*  
M. Beneke, F. Dighera, A. Hryczuk  
JHEP **1609** (2016) 031 [[arXiv:1607.03910 \[hep-ph\]](#)]
- [18] *Relic density of wino-like dark matter in the MSSM*  
M. Beneke, A. Bharucha, F. Dighera, C. Hellmann, A. Hryczuk, S. Recksiegel, P. Ruiz-Femenia  
JHEP **1603** (2016) 119 [[arXiv:1601.04718 \[hep-ph\]](#)]
- [19] *Relic density computations at NLO: infrared finiteness and thermal correction*  
M. Beneke, F. Dighera, A. Hryczuk  
JHEP **1410** (2014) 045 [[arXiv:1409.3049 \[hep-ph\]](#)]
- [20] *Indirect Detection Analysis: Wino Dark Matter Case Study*  
I. Cholis, A. Hryczuk, R. Iengo, P. Ullio and M. Tavakoli  
JCAP **1407** (2014) 031 [[arXiv:1401.6212 \[astro-ph.HE\]](#)]
- [21] *The one-loop and Sommerfeld electroweak corrections to the Wino dark matter annihilation*  
A. Hryczuk, R. Iengo  
JHEP **1201** (2012) 163, Erratum-ibid. **1206** (2012) 137 [[arXiv:1111.2916 \[hep-ph\]](#)]
- [22] *The Sommerfeld enhancement for scalar particles and application to sfermion co-annihilation regions*  
A. Hryczuk  
Phys. Lett. **B699** (2011) 271-275, [[arXiv:1102.4295 \[hep-ph\]](#)]



- [23] *Relic densities including Sommerfeld enhancements in the MSSM*  
A. Hryczuk, R. Iengo, P. Ullio  
JHEP **1103** (2011) 069, [[arXiv:1010.2172 \[hep-ph\]](#)]

### Contributing Author Publications & Proceedings

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- [C1] *Dark Matter Freeze-out and Freeze-in beyond Kinetic Equilibrium*  
A. Hryczuk  
LHEP 2023 (2023) 344
- [C2] *The Forward Physics Facility at the High-Luminosity LHC*  
A. Hryczuk in Feng et al.  
J.Phys.G 50 (2023) 3, 030501 [[arXiv:2203.05090 \[hep-ex\]](#)]
- [C3] *Non-equilibrium Effects in the Evolution of Dark Matter*  
A. Hryczuk  
Acta Phys. Pol. B Proc. Suppl. 13, 725 (2020)
- [C4] *Forbidden Freeze-in*  
L. Darme, A. Hryczuk, D. Karamitros, L. Roszkowski  
Acta Phys. Pol. B Proc. Suppl. 13, 733 (2020)
- [C5] *Phase transitions and gravitational waves in models of  $\mathbb{Z}_N$  scalar dark matter*  
N. Benincasa, K. Kannike, A. Hektor, A. Hryczuk, K. Loos  
PoS EPS-HEP2019 (2020) 089
- [C6] *Dark Matter Relic Density Revisited: The Case For Early Kinetic Decoupling*  
T. Binder, T. Bringmann, M. Gustafsson, A. Hryczuk  
Contribution to Moriond EW 2018 [[arXiv:1805.00526 \[hep-ph\]](#)]
- [C7] *Science with e-ASTROGAM (A space mission for MeV-GeV gamma-ray astrophysics)*  
A. Hryczuk in A. De Angelis et al.  
JHEAp 19 (2018) 1-106 [[arXiv:1711.01265 \[astro-ph.HE\]](#)]
- [C8] *Optical Search for QED Vacuum Magnetic Birefringence, Axions and Photon Regeneration*  
A. Hryczuk in P. Pugnati et al.  
CERN-SPSC-2006-035 (Nov 2006)